

Introduction:

Antibiotic resistance is a major global health threat that makes infections harder to treat, especially with pathogens like MRSA that can resist multiple drugs. Staphylococcus is a genus of bacteria commonly found on the skin and in the nose, and some species, such as Staphylococcus aureus, can cause serious infections when they enter the body.

This increases the need for stronger or more toxic treatments and raises the risk of complications in medical procedures such as surgery and chemotherapy.

At the same time, the discovery of new antibiotics had slowed, while resistance continues

to grow, creating a gap between need and development. As a result, researchers are turning

to alternative sources of antimicrobial compounds. Natural microorganisms, including Streptomyces species and fungi, have historically produced many antibiotics through compounds used to compete with other microbes. However, some microbial diversity remains unexplored, suggesting that many useful antimicrobial agents are still undiscovered.

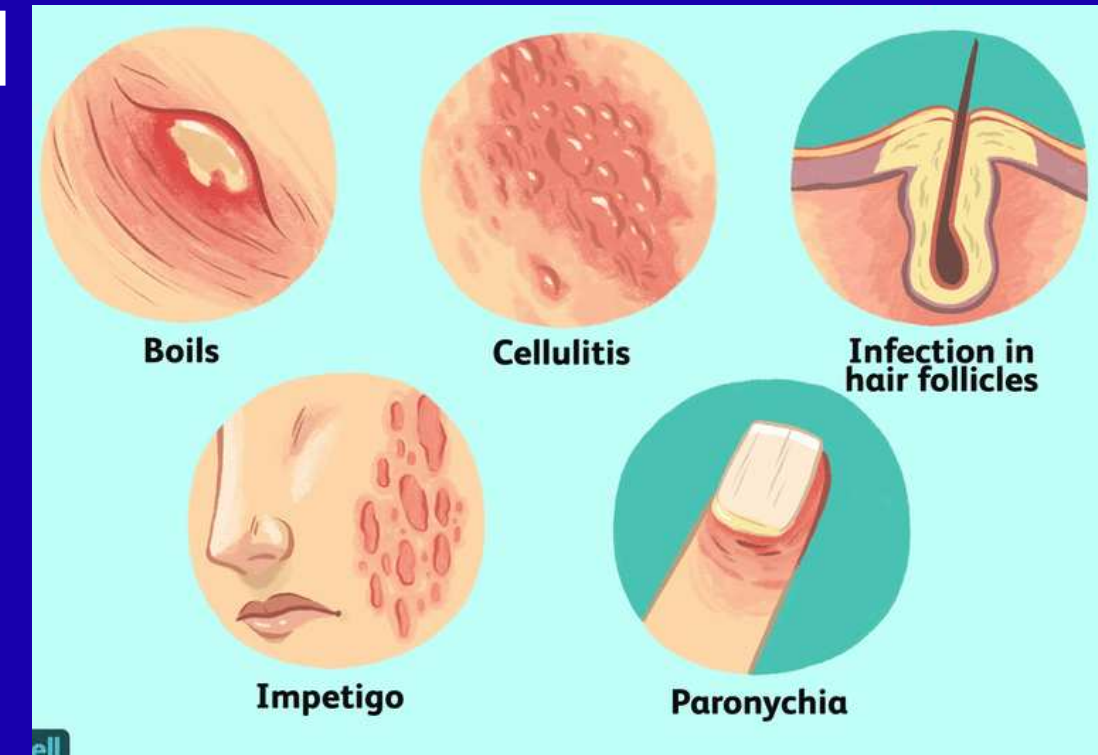


Figure 1. Staphylococcus Skin Infections (Iannelli, 2025).

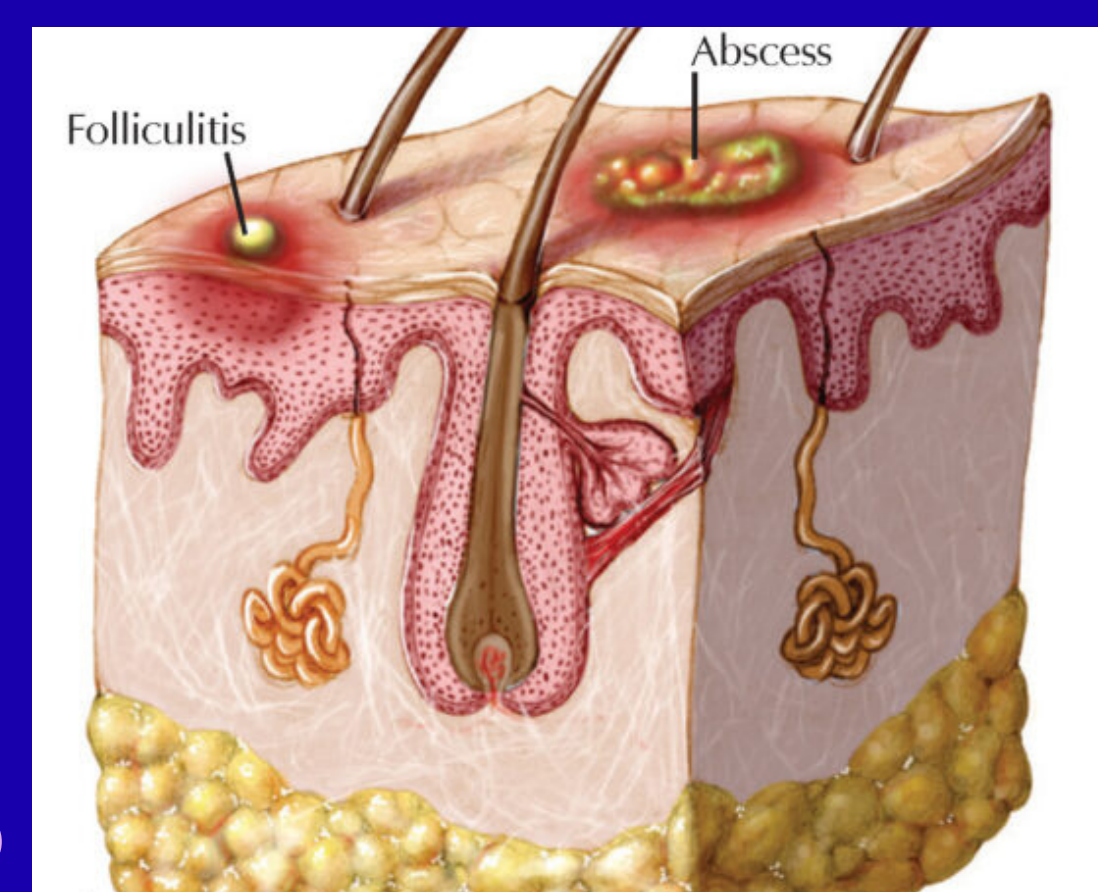


Figure 2. Cross Section of skin layers with MRSA Infection (Akins, 2021).

Purpose:

This study modeled early-stage antibiotic discovery by investigating the antibacterial activity of crude extracts derived from fungal and Streptomyces isolates. The effectiveness of these extracts was tested against bacteria, including resistant strains such as MRSA, to evaluate their potential as sources of new antimicrobial compounds.

Evaluation of Antibacterial Efficacy of Fungal Extracts Against Antibiotic-Resistant Bacteria

Methods:

Culture Growth

Fungal and Streptomyces isolates were grown on **agar plates and then transferred to liquid media for metabolite production**. Cultures were incubated with shaking for several days.



Figure 3. Culture plates placed under incubation (Photo taken by Researcher).

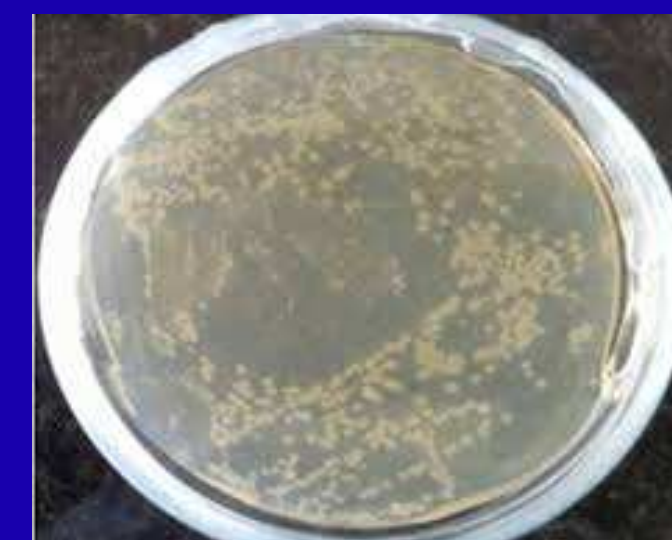


Figure 4. E.coli grown on agar plate (Photo taken by Researcher).

Disk Diffusion Assay

Test bacteria were spread onto Mueller-Hinton agar plates. Filter paper disks loaded with extract solutions were placed on the agar. **Solvent controls and antibiotic controls were included.** Plates were incubated for 16–20 hours.



Figure 7. MHA liquified ready to be poured into petri-dishes (Photo taken by Researcher).



Figure 8. E.coli (control) spread evenly to create a consistent bacterial lawn (Photo taken by Researcher).

Extraction

Bioactive compounds were extracted using two methods: **ethanol maceration of dried biomass and liquid-liquid extraction** of culture supernatant with ethyl acetate. Solvents were evaporated, and extracts were reconstituted to known concentrations.



Figure 5. The Ethyl Acetate (Photo taken by Researcher).



Figure 6. Process of Liquid-Liquid Extraction (Photo taken by Researcher).

Data Collection

Zones of inhibition were **measured in millimeters** to determine the level of antibacterial activity. Each extract was tested in triplicate, and **average inhibition values were calculated** to ensure accuracy and reliability of the results.



Figure 9. After Incubation of Extract A (Streptomyces-ethanol) (Photo taken by Researcher).



Figure 10. After Incubation of Extract B (Streptomyces-Ethyl Acetate) (Photo taken by Researcher).

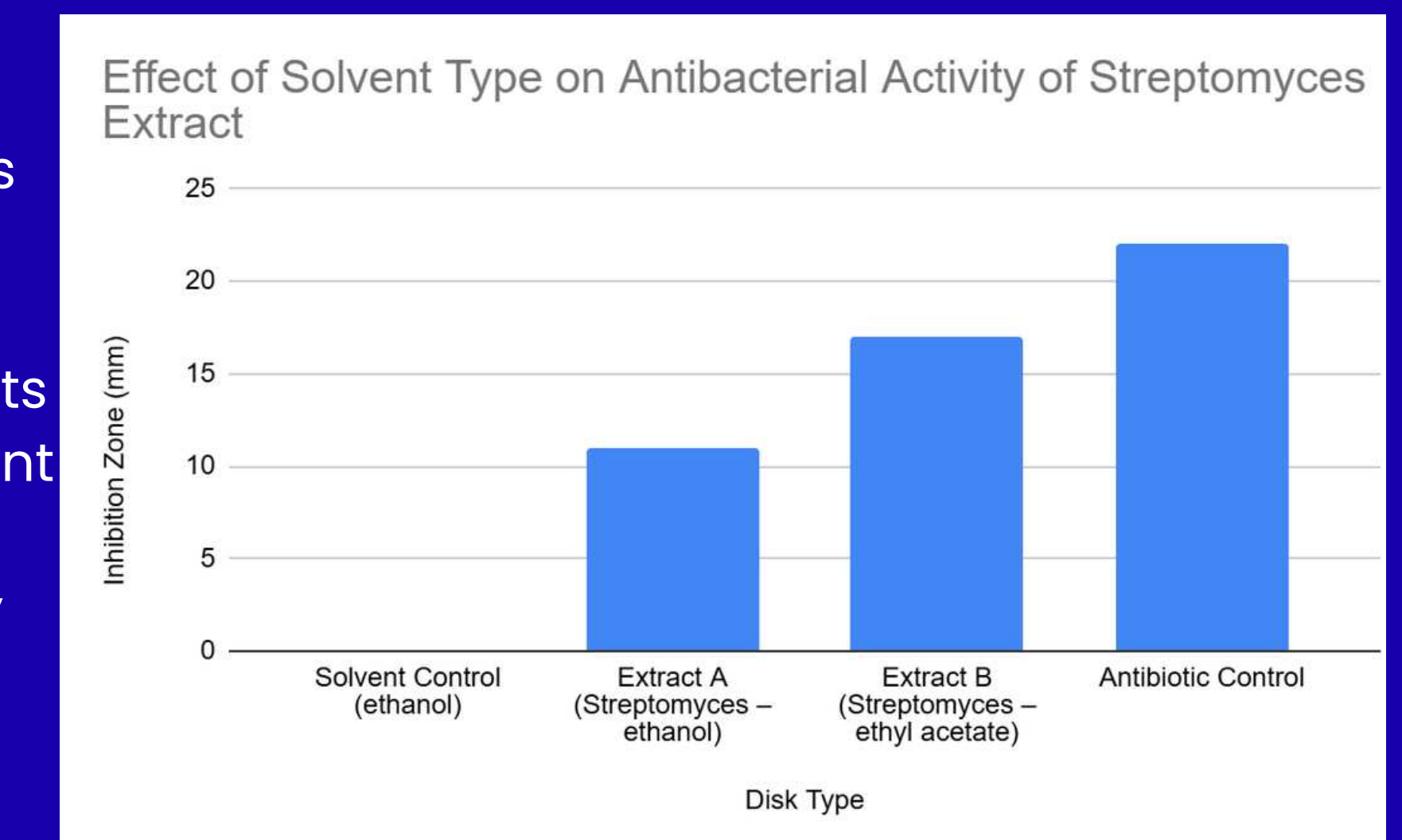
Results:

Disk Type	Inhibition Zone (mm)	Interpretation
Solvent Control (ethanol)	0 mm	No inhibition
Extract A (Streptomyces – ethanol)	11 mm	Moderate inhibition
Extract B (Streptomyces – ethyl acetate)	17 mm	Stronger inhibition
Antibiotic Control	22 mm	Strong inhibition

Figure 11. Modeled disk diffusion results showing inhibition zones for Streptomyces extracts and control treatments in table format with interpretations.

(Table created by Researcher using Google Sheets).

Figure 12. Modeled disk diffusion results showing inhibition zones for Streptomyces extracts prepared with different solvents. (Graph created by Researcher using Google Sheets).



Conclusions:

The control trial using E. coli confirmed that the disk diffusion method was reliable, as shown by consistent antibiotic inhibition (22 mm average) and no inhibition from the solvent control. This validation ensures that any observed antibacterial effects are due to the tested compounds rather than experimental error. Overall, this study establishes a reliable testing framework and demonstrates how extraction methods influence antibacterial outcomes. This brings the project closer to its goal by enabling future screening of natural compounds that may contribute to the development of treatments for antibiotic-resistant bacteria.

Citations:

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- MicrobeOnline. (2020). Personal protective equipment (PPE) for laboratory work. Retrieved from <https://microbeonline.com/personal-protective-equipment-ppe/>
- SICWeb. (2021). Medical labs and general laboratory equipment maintenance tips. Retrieved from <https://sicweb.com/blog/109-medical-labs-and-general-laboratory-equipment-maintenance-tips>
- Singh, D., & Kumar, R. (2018). Graph showing the mean inhibition zone diameter produced by each concentration of fungal extract. ResearchGate. https://www.researchgate.net/figure/fig-2-Graph-showing-the-mean-inhibition-zone-diameter-produced-by-each-concentration-of_fig1_322329937
- Zong, Y., & Zhang, X. (2020). Evaluation of the antibacterial efficacy of fungal extracts against drug-resistant bacteria. Nature. <https://doi.org/10.1038/s41598-020-74262-y>